Chemistry - Undergraduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following metal ions cannot be used as a paramagnetic quencher?
   - (a) Ti_3+
   - (b) Cr_3+
   - (c) Fe_3+
   - (d) Zn_2+
2. What is the orbital angular momentum quantum number, l, of the electron that is most easily removed when ground-state aluminum is ionized?
   - (a) 3
   - (b) 2
   - (c) 1
   - (d) 0
3. Of the following ions, which has the smallest radius?
   - (a) K+
   - (b) Ca_2+
   - (c) Sc_3+
   - (d) Rb+
4. Which of the following types of spectroscopy is a light-scattering technique?
   - (a) Nuclear magnetic resonance
   - (b) Infrared
   - (c) Raman
   - (d) Ultraviolet-visible
5. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
   - (a) all molecular bonds absorb IR radiation
   - (b) IR peak intensities are related to molecular mass
   - (c) most organic functional groups absorb in a characteristic region of the IR spectrum
   - (d) each element absorbs at a characteristic wavelength

True / False (10 marks)

Answer True or False

6. True or False: The atomic radii of the lanthanide elements decrease across the period from La to Lu due to increasing nuclear charge.
7. True or False: Neither position nor momentum of a quantum mechanical particle in a one-dimensional box can be known exactly.
8. True or False: A magnetic field of 0.5 T is sufficient to cause significant polarization in 13 C at 298 K.
9. True or False: In a three-component mixture, a maximum of five phases can be in equilibrium with each other.
10. True or False: NaCl has a greater lattice enthalpy than MgO because sodium and chloride ions are more stable than magnesium and oxide ions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the R-branch in the vibrational-rotational spectrum of a diatomic molecule, focusing on the specific transitions involved and their implications for molecular behavior.
12. Calculate the NMR frequency of 31 P in a 20.0 T magnetic field and discuss the factors that influence this frequency in nuclear magnetic resonance spectroscopy.

End of Paper